

Deterministic-Numerator Analytic-Tail Certificate

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Abstract

RH-251 left the deterministic numerator tail as a separate obligation in the finite-head/analytic-tail interface. RH-46 supplies a stronger analytic input than the finite coefficient dictionary alone: the deterministic one-step numerator G is holomorphic and nonzero on $|z| < \lambda$, where $\lambda = 1.6785735104283177\dots$. After Hardy scaling by $r_H = 0.85$, the logarithm of $G(z/r_H)$ therefore has radius at least $\rho_* = r_H\lambda = 1.42678748386407\dots$. We prove an all-order Cauchy tail bound for the target coefficients and its exponential determinant conversion. The boundary supremum entering the bound is not numerically certified, so no cloud or quotient uniformity is claimed.

1 Input and scope

Write the deterministic factor from RH-46 as

$$\hat{D}_{0,\text{bulk},2}(z) = \frac{G(z)}{1 - z^2/\lambda}, \quad G(0) = 1, \quad (1)$$

where G is holomorphic and nonzero on $|z| < \lambda$ [2]. RH-243 identifies the one-step Hardy-scaled trace-style target by

$$\log G_H(z) = - \sum_{n \geq 2} \frac{a_n}{n} z^n, \quad G_H(z) := G(z/r_H), \quad r_H = 0.85. \quad (2)$$

The coefficients a_n are deterministic target coefficients, not yet the coefficients of a selected noisy cloud [1]. We set $a_1 := 0$, so coefficient estimates below may be stated uniformly for $n \geq 1$.

Proposition 1.1 (Scaled zero-free disk). *The function G_H is holomorphic and nonzero on*

$$|z| < \rho_* := r_H\lambda = 1.42678748386407\dots > 1. \quad (3)$$

Consequently there is a unique holomorphic logarithm L_H on this disk with $L_H(0) = 0$, and $L_H = \log G_H$ in (2).

Proof. The substitution $z \mapsto z/r_H$ maps $|z| < r_H\lambda$ into $|z| < \lambda$. Holomorphicity and nonvanishing are preserved. The disk is simply connected and $G_H(0) = 1$, so the normalized logarithm exists and is unique. $\square$

2 The all-order tail theorem

Definition 2.1 (Boundary budget). For $0 \leq R < S < \rho_*$ define

$$M_S := \max_{|z|=S} |L_H(z)|, \quad \Phi_N(R, S) := \frac{(R/S)^N}{1 - R/S}, \quad N \geq 1. \quad (4)$$

The first omitted order is N ; thus the tail starts at $n = N$.

Theorem 2.2 (Analytic deterministic-target tail). *For every $0 \leq R < S < \rho_*$ and every integer $N \geq 1$,*

$$\frac{|a_n|}{n} \leq M_S S^{-n} \quad (n \geq 1), \quad \sum_{n \geq N} \frac{|a_n| R^n}{n} \leq M_S \Phi_N(R, S). \quad (5)$$

In particular, since $\rho_ > 1$, the all-order deterministic target tail on the unit disk exists for every $1 < S < \rho_*$.*

Proof. By Cauchy's coefficient estimate applied to the holomorphic function L_H on $|z| < S$, the coefficient of z^n in L_H has modulus at most $M_S S^{-n}$. Equation (2) says that this coefficient is $-a_n/n$. Multiplying by R^n and summing the geometric series proves (5). $\square$

Corollary 2.3 (Determinant conversion). *If $E_N(R, S) := M_S \Phi_N(R, S)$, then truncating the target logarithm after order $N - 1$ incurs relative multiplicative error at most*

$$\left| \exp \left(- \sum_{n \geq N} a_n z^n / n \right) - 1 \right| \leq e^{E_N(R, S)} - 1, \quad |z| \leq R. \quad (6)$$

Proof. Theorem 2.2 bounds the modulus of the omitted logarithm by $E_N(R, S)$. The elementary inequality $|e^w - 1| \leq e^{|w|} - 1$ gives the claim. $\square$

Remark 2.4 (What the theorem does not provide). The theorem is an exact analytic interface, but M_S is an unknown boundary constant. A finite evaluation of the order-2–12 polynomial cannot bound the full L_H on the same circle. Thus the result is not a numerical uniform tail certificate for the moving cloud and does not invoke RH-240.

3 Finite diagnostic

We read the archived RH-243 coefficients through order twelve and evaluate the truncated logarithm on 4096 equally spaced points for $S \in \{1.05, 1.15, 1.25, 1.35\}$. These values are diagnostics only; the exact quantity M_S remains unbounded by the experiment. The geometric factor for the unit-disk tail beginning at order 13 is shown below.

S	truncated max $ L_H $	$\Phi_{13}(1, S)$	status
1.05	0.579287	11.136748	diagnostic only
1.15	0.797017	1.246048	diagnostic only
1.25	1.106719	0.274878	diagnostic only
1.35	1.559934	0.077970	diagnostic only

The last column is essential: multiplying the displayed truncated supremum by Φ_{13} is not a proof of the tail bound. It only illustrates the radius budget once a rigorous M_S is supplied.

4 Gate and route audit

The new exact statement closes the deterministic-target analytic-tail existence obligation on every strict subdisk of radius ρ_* . It does not identify the current noisy cloud with a_n , does not supply a uniform quotient block estimate, and does not produce a finite numerical value for M_S . Therefore Gates A–E remain false/open. No Hilbert–Polya operator, Riemann zero identification, zeta-divisor equality, or RH implication is claimed.

The next result should either interval-certify M_S on a useful circle or provide a genuinely expanded resolved candidate window. Reweighting the frozen RH-248 shell class would not be a new input.

References

- [1] Bin Wang. A deterministic numerator coefficient-anchor dictionary. *Preprint*, 2026. RH-243 preprint.
- [2] Bin Wang. Small-noise mesh double pole. *Preprint*, 2026. RH-46 preprint.